

Naloga 20.1. Izračunajte.

- | | | | |
|-------------------|--------------------------|----------------------------|----------------|
| • $\sqrt{-4}$ | • $\sqrt{-2}$ | • $\sqrt{\frac{10}{-100}}$ | • $\sqrt{-20}$ |
| • $\sqrt{-256}$ | • $\sqrt{-200}$ | • $\sqrt{-0.75}$ | • $\sqrt{-63}$ |
| • $\sqrt{-10000}$ | • $\sqrt{-\frac{3}{16}}$ | • $\sqrt{-250}$ | • $\sqrt{-64}$ |

Naloga 20.2. Zapišite realno in imaginarno komponento danega števila. Določite, ali je število realno ali imaginarno.

- | | | |
|-----------------|------------------------------------|------------------------------|
| • $z = 2 - 4i$ | • $w = i$ | • $w = \frac{1}{5}$ |
| • $w = -1 + 3i$ | • $z = \sqrt{2} + \sqrt{3}i$ | • $z = \frac{\sqrt{5}}{2}i$ |
| • $z = -4i$ | • $w = \frac{1}{2} + \frac{1}{3}i$ | • $w = \sqrt{13} - \sqrt{5}$ |
| • $w = 12$ | • $z = \frac{\sqrt{3}}{2}i - 2$ | |
| • $u = 2i - 12$ | | |

Naloga 20.3. V kompleksni ravnini upodobite dana števila.

- | | | |
|-----------------|------------------|------------------------------|
| • $z = -i$ | • $w = 2 + 0.5i$ | • $w = -3i$ |
| • $w = 2 - 4i$ | • $w = 4$ | • $z = 2.5 + 1.75i$ |
| • $z = -2i + 3$ | • $u = \sqrt{3}$ | • $w = \sqrt{2} - \sqrt{3}i$ |

Naloga 20.4. V kompleksni ravnini upodobite množico točk, ki ustreza danim pogojem.

- $(\operatorname{Re}(z) = 3) \wedge (\operatorname{Im}(z) > -1)$
- $(\operatorname{Re}(z) \leq 1) \vee (\operatorname{Im}(z) = -2)$
- $\operatorname{Im}(z) = 2 + 3\operatorname{Re}(z)$
- $(\operatorname{Re}(z) \geq -3) \wedge (1 < \operatorname{Im}(z) \leq 2)$
- $(|\operatorname{Re}(z)| \leq 1) \vee (\operatorname{Im}(z) = 2)$
- $(\operatorname{Re}(z) \leq 1) \wedge (\operatorname{Im}(z) = -2)$
- $(\operatorname{Re}(z) \leq \operatorname{Im}(z)) \wedge (\operatorname{Im}(z) = -2)$
- $(\operatorname{Re}(z) = -1) \vee (\operatorname{Im}(z) = 3)$
- $(\operatorname{Re}(z) = 2) \wedge (\operatorname{Im}(z) = 1)$
- $(\operatorname{Re}(z) < 3) \vee (\operatorname{Im}(z) \geq -2)$
- $(\operatorname{Re}(z) = 2\operatorname{Im}(z) + 1) \wedge (|\operatorname{Im}(z)| = 1)$

Naloga 20.5. Zapišite kompleksno število, za katerega veljajo zapisani pogoji.

- $\operatorname{Re}(z) + 2\operatorname{Im}(z) = 3 \quad \wedge \quad \operatorname{Re}(z) = (\operatorname{Im}(z))^2$
- $\operatorname{Im}(z) = 2\operatorname{Re}(z) \quad \wedge \quad \operatorname{Re}(z)\operatorname{Im}(z) = 8$
- $\operatorname{Im}(z) + 2\operatorname{Re}(z) = 13 \quad \wedge \quad \operatorname{Im}(z) - \operatorname{Re}(z) = -2$
- $5\operatorname{Im}(z) + 2\operatorname{Re}(z) = -2 \quad \wedge \quad \operatorname{Re}(z) - 3\operatorname{Im}(z) = -1$
- $\frac{1}{2}\operatorname{Re}(z) - \operatorname{Im}(z) = 3 \quad \wedge \quad \operatorname{Im}(z) + \operatorname{Re}(z) = -2$
- $0.2\operatorname{Im}(z) + 0.5\operatorname{Re}(z) = 2 \quad \wedge \quad 1.5\operatorname{Re}(z) - 0.4\operatorname{Im}(z) = 1$
- $\operatorname{Re}(z) - \operatorname{Im}(z) = -1 \quad \wedge \quad 0.6\operatorname{Re}(z) + 0.4\operatorname{Im}(z) = 7$
- $3\operatorname{Im}(z) + 2\operatorname{Re}(z) = 3 \quad \wedge \quad 3\operatorname{Im}(z) - 4\operatorname{Re}(z) = 0$
- $3\operatorname{Re}(z) - 2\operatorname{Im}(z) = 2 \quad \wedge \quad 2\operatorname{Re}(z) - 3\operatorname{Im}(z) = -2$